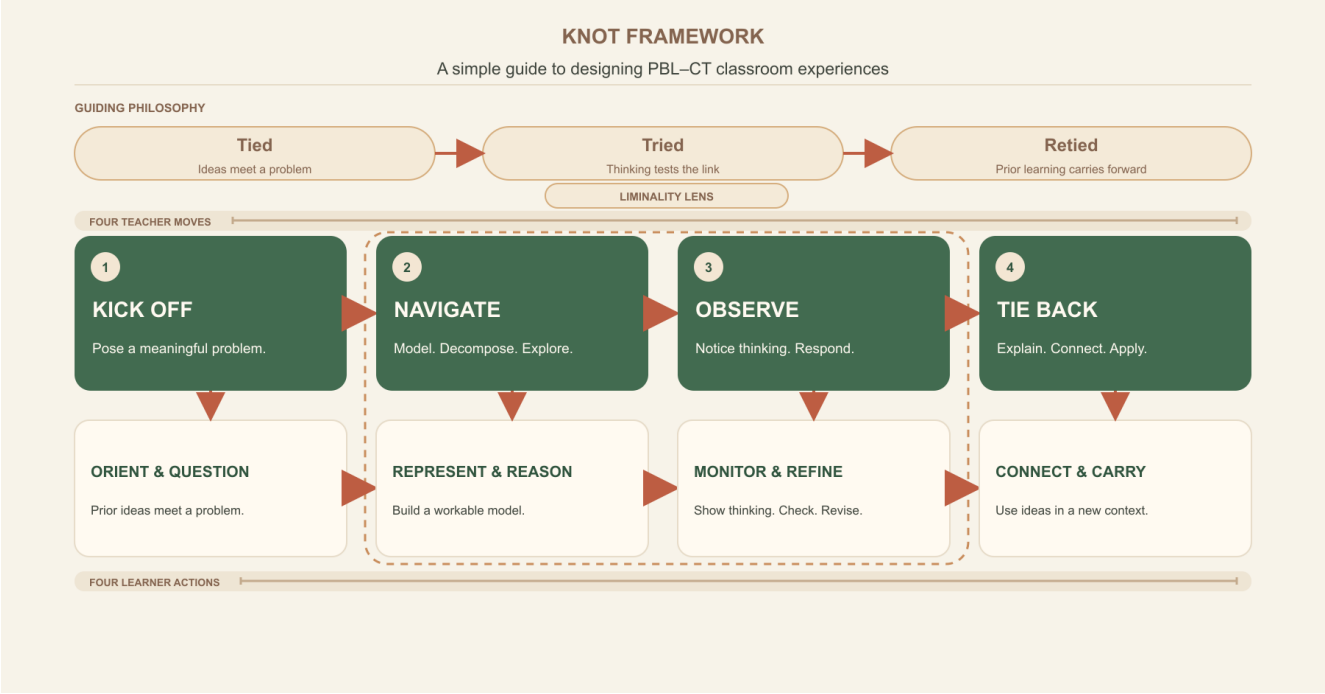
KNOTTY CLASS

FRAMEWORK AND PHILOSOPHY GUIDE

A self-contained design guide for teachers and AI tools

KNOT is Not One-time Transformation. The name is recursive by design: what learners understand can shape a new question and lead to further learning.



Tied, Tried, Retied

Learning connects ideas to a problem, tests those connections through thinking and evidence, and carries or reshapes understanding in what comes next.

This guide presents the KNOT framework developed through the research work of Prakash Hegade. Read the complete research at prakashbh.github.io/KNOT.

1. How to use this guide

This PDF gives an AI tool the KNOT philosophy, framework, design principles, and guidance for producing a case study with an initial class plan. The separate KNOTTY Class Design Brief supplies six details: subject or course, concept or topic, student level or year, approximate number of students, class duration, and classroom format. Upload both documents together.

STEP	WHAT TO DO
1. Complete the short Design Brief	Provide only the subject or course, concept or topic, student level or year, approximate number of students, class duration, and classroom format.
2. Upload both documents	Upload this KNOT Guide and the completed Design Brief to the LLM. The guide defines KNOT; the brief defines this class.
3. Request the first design	The short prompt in the Design Brief asks the LLM to prepare a class plan. This guide explains that the first output should include a student-facing case study and a teacher-facing KNOT plan, with suggested learning outcomes.
4. Review and refine	Treat the first output as a starting design. Check subject accuracy, timing, suitability for your learners, and the proposed outcomes. Refine it for your class before teaching.

What KNOT means

KNOT is Not One-time Transformation. This is a recursive definition. Learning does not end with one change in understanding. What a learner constructs can shape the next question, be tested in a new situation, and be reconstructed through further inquiry and application.

What the philosophy means

Tied, Tried, Retied describes the learning philosophy behind the framework. Learners tie ideas to a meaningful problem, try those connections through thinking and evidence, then retie them as they carry or reshape understanding in another context. A knot is a connection that can be tested, loosened, strengthened, and made useful in what comes next.

Purpose of the framework

KNOT is a practice-oriented, evidence-informed framework for designing PBL-CT classroom experiences that support knowledge construction. It helps teachers plan a purposeful beginning, student reasoning, attention to evidence of thinking, and an opportunity to connect learning beyond the immediate task. It offers design guidance, not a guaranteed outcome for every learner.

2. The four KNOT teacher moves

Use the moves as connected planning prompts. The emphasis and order can vary with the concept, the learners, and the class. A short activity may use each move briefly; a larger case or project may return to them over several lessons.

TEACHER MOVE	PLAN FOR	LEARNER ACTIONS	EVIDENCE TO NOTICE
Kick off	A meaningful problem, case, question, or challenge that connects to the learning objective and gives students a reason to begin.	Orient to the situation, connect prior ideas, identify what is known and unknown, and ask questions.	Questions, predictions, initial representations, prior knowledge, and assumptions.
Navigate	Tasks, resources, prompts, examples, dialogue, and scaffolded choices that help students work through the problem.	Represent, model, decompose, identify patterns, explore options, reason, and develop possible responses.	Diagrams, traces, explanations, decisions, drafts, strategies, peer discussion, and help-seeking.
Observe	Planned opportunities to inspect student thinking and respond with feedback, a prompt, a resource, more practice, or a new challenge.	Make thinking visible, compare it with evidence, notice uncertainty, and check or revise an approach.	What students can explain, where they hesitate, what errors recur, and how their reasoning changes.
Tie back	A close or follow-up task that asks students to explain, connect, apply, reflect, or carry an idea into a changed context.	Summarise what they understand, justify a connection, apply an idea, and identify a next question.	Explanations, transfer attempts, reflections, solutions, and links to the learning objectives.

Observe continues throughout the class. Plan a visible check or decision point, while also noticing student thinking during the other moves. Use what students show to decide what support or opportunity should come next.

What flexible use looks like

- The teacher may return to Navigate after observing that students need another representation or example.
- Tie back may lead to a new Kick off in a later class when an application raises another question.
- A lesson does not need to force all four moves into equal time or treat them as a fixed sequence.
- The class should still have a clear learning purpose and a way to see what students have learned.

3. Learning dimensions and knowledge construction

The framework can support three related learning dimensions: self-directed learning, problem solving, and transfer of knowledge. Knowledge construction is the broader process that these dimensions can help make visible. Teachers may prioritise one or more dimensions for a particular class. Do not force every class to give all dimensions equal attention.

DIMENSION	WHAT IT INVOLVES	POSSIBLE EVIDENCE
Self-directed learning	Taking initiative and responsibility for parts of learning. This can include identifying a learning need, selecting or using resources, practising, monitoring progress, and seeking help strategically. It does not mean learning without teacher support.	Student questions, planning choices, self-checks, independent practice, resource use, purposeful help-seeking, and reflection.
Problem solving	Understanding and representing a problem, deciding what information matters, decomposing it, considering constraints, comparing approaches, and explaining or checking a solution.	Models, diagrams, traces, justified choices, working solutions, comparisons, explanations, and revisions.
Transfer of knowledge	Recognising when an idea from one lesson or context can help in another, then adapting and explaining its use. A familiar story alone does not demonstrate transfer.	A student identifies a relevant concept, explains why it fits the new context, and shows how it was adapted or applied.
Knowledge construction	The wider process of making, organising, connecting, testing, explaining, and revising understanding. It can be seen across the other dimensions rather than as a separate fifth classroom activity.	Connected concepts, more developed explanations, reasoned decisions, revisions, and applications that show what the learner now understands.

Use evidence carefully

An activity creates an opportunity for learning; it does not by itself prove that learning occurred. Plan how students will show their thinking. Read that evidence in relation to the stated objective, and avoid claiming a result that the evidence does not support. A student may demonstrate one dimension clearly while another remains developing.

Possible alignment check

For each learning objective, identify at least one student task and one piece of evidence that could show progress. If an objective has no matching activity or evidence, revise the plan before teaching.

4. Design foundations and useful lenses

KNOT draws together ideas that help teachers design and interpret learning. Use each in ways that serve the concept and objective. They are resources for thinking, not a list of techniques to include in every class.

IDEA	HOW IT CAN INFORM A CLASS
Problem-Based Learning (PBL)	A problem, case, or challenge gives learners a reason to investigate. The teacher guides inquiry and provides resources or scaffolding while students do meaningful work on the problem.
Computational Thinking (CT)	Students can formulate or represent a problem, decompose it, notice patterns, abstract relevant features, develop a procedure, and evaluate a solution. Select the practices that fit the topic; do not force every CT term into every lesson.
Reflective inquiry	Students examine a question, consider evidence, test an explanation, and reflect on what their attempt shows. Give them opportunities to explain why a solution or representation makes sense.
Liminality lens	Some learners may experience uncertainty or partial understanding while working with difficult ideas. Treat liminality as an interpretive lens for possible transitions, not a fixed stage every student passes through or a label for every difficulty.
Scaffolding, dialogue, and feedback	Examples, prompts, peer discussion, practice, and timely feedback can help students continue. Choose support in response to what students show, and adjust it as their needs change.

Assessment belongs inside the design

Assessment is not only a final score. A prompt, worked example, group explanation, trace, draft, or exit response can provide evidence while students are learning. A plan should state what the teacher will notice, what criteria matter, and what response may follow. Where a formal assessment is required, align it with the learning objectives and the work students practised.

> **Use support language: the class is designed to create opportunities for self-directed learning, problem solving, transfer, and knowledge construction. Do not claim that a lesson guarantees these outcomes or that the framework alone caused them.**

5. Building a useful case study

A case study should give students something meaningful to investigate or decide. A story can create interest, but the learning comes from the problem students have to interpret, the choices they must make, and the concepts they need to use. Keep the case within the time, level, and resources stated in the Design Brief.

A strong student-facing case usually includes

PART	WHAT TO INCLUDE
1. Title and hook	A concise title and an opening situation that gives students a reason to care or wonder.
2. Situation and role	Who is involved, what has happened, and what role students are asked to take. Give only information students need to begin.
3. The central challenge	A clear question, decision, investigation, or design problem linked to the learning objectives.
4. Evidence and constraints	The facts, data, diagrams, conditions, limits, or assumptions students can use. Distinguish supplied facts from hypothetical assumptions.
5. Student task	What students must produce or explain, such as a model, algorithm, recommendation, comparison, diagram, written rationale, or presentation.
6. Inquiry prompts	Questions that help students interpret, represent, decompose, compare, test, justify, or reflect without giving away the whole solution.

Teacher-facing case notes

- Identify the target concepts and how they relate to the class objectives.
- Offer more than one plausible route when the problem allows it. Explain what makes a response well reasoned.
- Anticipate likely misconceptions, missing prerequisites, and moments where a prompt or example may help.
- Suggest what evidence to collect while students work and what a useful Tie back could ask.
- Keep any suggested solution or teacher-only guidance separate from the student-facing case.

Accuracy and context

Use data, policies, technical claims, and local details supplied by the teacher. If an authentic case needs current or specialised facts, ask for sources or mark the details that require verification. If inventing a scenario for learning, label it as fictional and keep its assumptions internally consistent. Never present invented statistics or events as real.

6. Turning the case into a KNOT class

The case and lesson plan should work together. The case creates the student-facing challenge; the plan shows how the teacher will help learners engage with it, work through it, and make their learning visible.

Expected lesson-plan structure

PLAN ELEMENT	WHAT THE LLM SHOULD PROVIDE
Class snapshot	Topic, level, duration, assumed prior knowledge, mode of work, and the objectives copied or carefully refined from the brief.
Case handout	A polished student-facing case with the situation, challenge, available evidence, constraints, deliverable, and inquiry prompts.
Timed sequence	A practical sequence organised around Kick off, Navigate, Observe, and Tie back. Total minutes must equal the class duration.
Teacher and learner actions	For each phase, state what the teacher does and what students do. Include key questions, instructions, materials, and grouping.
Evidence and response	Identify what the teacher will observe, how progress will be checked, and what feedback, support, or adjustment may follow.
Assessment alignment	Map each objective to a student activity and evidence of learning. Include success criteria or a short rubric when useful.
Teacher notes	List assumptions, possible misconceptions, differentiation or access suggestions, extensions, and an optional teacher-only solution direction.

Time and realism checks

- Make the total duration exact. Include transitions, instructions, group formation, and reporting time.
- Use only resources named in the brief, or clearly label a suggested resource for the teacher to approve.
- Scale the task to the students and available class time. A large project should not be squeezed into a short lesson.
- Include coding only when the teacher requests it or the brief makes it necessary. If unclear, ask or state the assumption.
- Offer a shorter alternative or extension only when it helps the teacher adapt the lesson.

When objectives are missing

Do not silently invent the teacher's intended outcomes. Offer a small set of measurable draft objectives based on the topic, mark them as suggestions, and ask the teacher to confirm or revise them before treating them as final.

7. Teacher review before using the plan

An AI-generated plan is a first draft. The teacher knows the students, course requirements, and classroom conditions. Use this review to decide whether the case and plan are ready.

- **Purpose:** Is the case clearly connected to the stated topic and learning objectives?
- **Student work:** Do students have a genuine task that requires them to think, decide, explain, or create?
- **Reasoning:** Does the Navigate work give students useful ways to represent, investigate, and check ideas?
- **Support:** Are the prompts, resources, examples, and feedback appropriate for the learners?
- **Observation:** Is there visible evidence that will help the teacher recognise what students understand and where they need support?
- **Application:** Does Tie back invite explanation, connection, reflection, or use in another context?
- **Assessment:** Does every objective have a suitable activity and evidence of learning?
- **Practicality:** Do timing, grouping, materials, accessibility, and class size fit the real setting?
- **Accuracy:** Are technical details correct? Are assumptions, fictional details, and facts clearly distinguished?
- **Teacher judgement:** Have you adapted the plan so it fits your students and your purpose?

What this guide does and does not establish

KNOT offers a way to design classroom experiences and attend to student thinking. It can help teachers plan opportunities that relate to self-directed learning, problem solving, transfer, and knowledge construction. A lesson plan is not proof that these outcomes occurred. Teachers should gather and interpret student evidence in the context of their own course. The framework is intended to be adapted thoughtfully across subjects and learner groups.

The framework in one sentence

KNOT is a practice-oriented, evidence-informed framework that guides teachers through four flexible moves: Kick off, Navigate, Observe, and Tie back, to design PBL-CT classroom experiences that support knowledge construction.

8. What the first design should include

The six details in the Design Brief are intended to make it easy for a teacher to get started. Use them to prepare a useful first draft rather than asking the teacher to complete a long intake form. Make reasonable assumptions where needed, label them clearly, and ask a follow-up question only when missing information prevents a responsible first design. The teacher can refine the draft afterwards.

First-output structure

OUTPUT PART	WHAT TO INCLUDE
Class snapshot and learning outcomes	Summarise the six class details. Propose a small number of clear, observable learning outcomes suited to the topic and student level. Mark them as suggested outcomes for the teacher to review and refine.
Student-facing case study	Create an engaging, bounded situation with a role, central challenge, relevant information, constraints, student task, expected deliverable, and inquiry questions. Make the case specific to the concept. Label invented contexts or data as fictional.
KNOT lesson plan	Organise the class around Kick off, Navigate, Observe, and Tie back. Give realistic timing, teacher actions, learner actions, materials, questions, and evidence to notice. The time allocations must add up to the class duration.
Objective and assessment alignment	Show which activity addresses each suggested learning outcome and what student evidence may indicate progress. Include simple success criteria or a brief assessment approach.
Teacher notes	State key assumptions, likely misconceptions, useful scaffolds, and any optional adaptation or extension. Keep any teacher-only solution direction separate from the student case.

The first design is a starting point

KNOT provides the design structure and this guide provides the framework information. The teacher brings subject knowledge and knowledge of the learners, then decides what to keep, change, or expand. A teacher may refine the suggested learning outcomes, case context, activities, assessment, or timing to fit the course.

This guide presents KNOT as an evidence-informed way to design learning opportunities. It does not promise that a class or framework will produce identical results for every learner.

End of Framework and Philosophy Guide